

Azure Cloud Fundamentals and Data Pipeline using ADF

Objective

The main goal of this assignment is to learn basic Azure cloud services and understand how data can be moved from one location to another using Azure Data Factory.

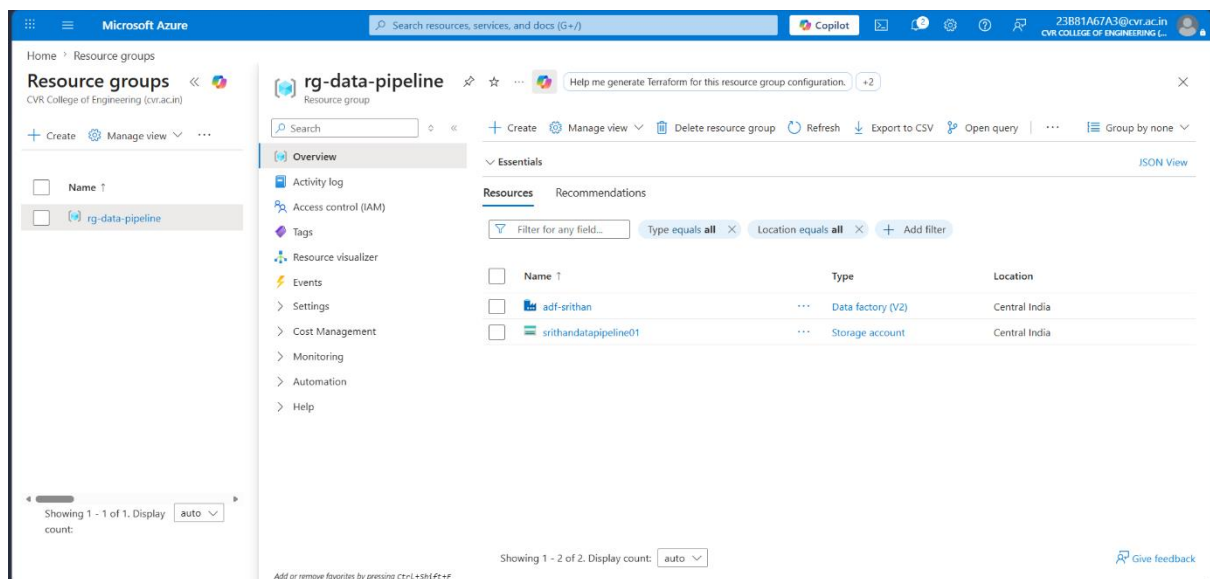
Services Used

- Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory
- IAM (Access Control)

What I Did

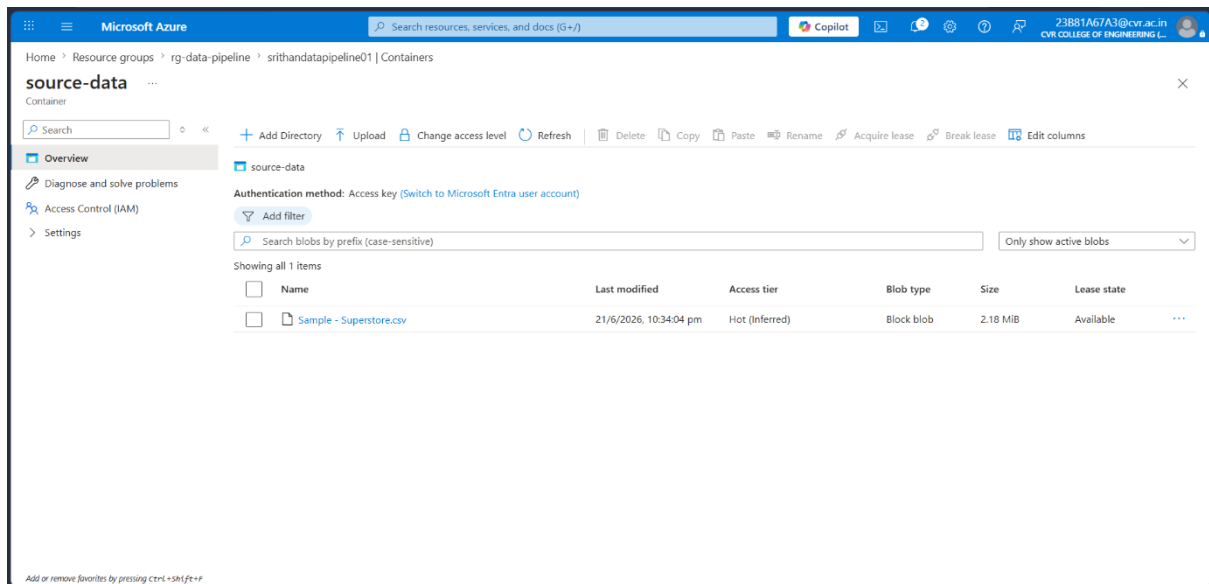
Resource Group

First, I explored the Azure portal and created a resource group called **rg-data-pipeline** to keep all the resources in one place.



Storage Setup

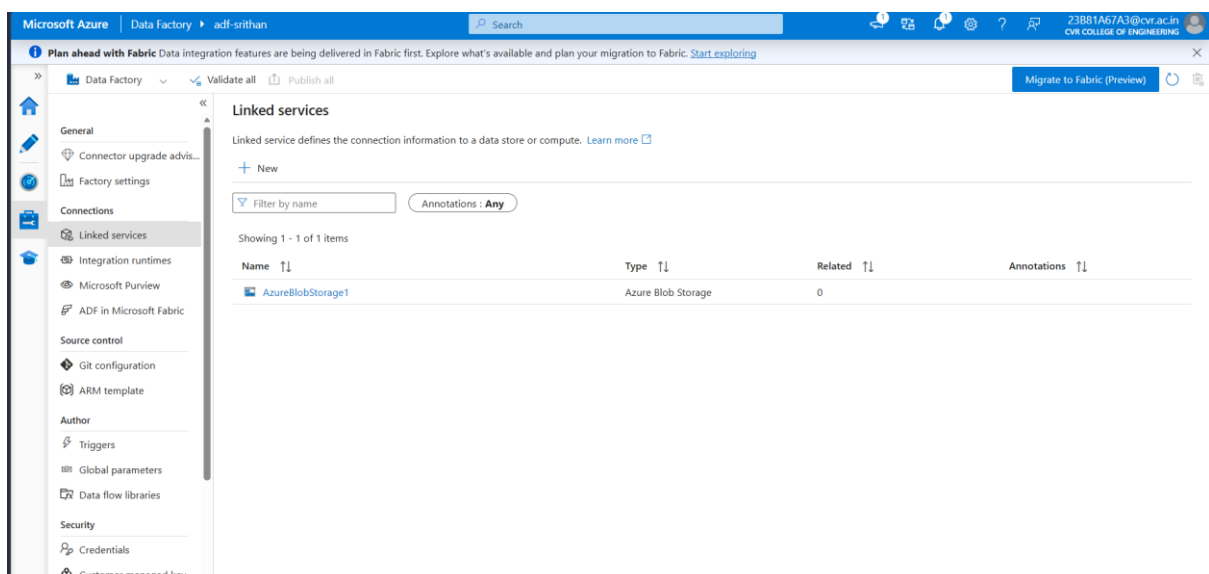
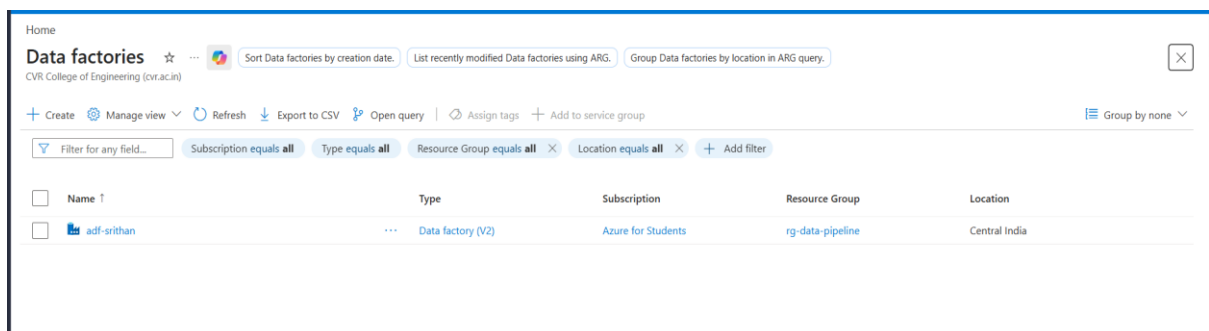
I created a storage account named **srithandatapipeline01**. Inside the storage account, I created a blob container called **source-data** and uploaded the **Sample-Superstore.csv** file.



Azure Data Factory

After that, I created an Azure Data Factory named **adf-srithan**. I explored the Author, Manage and Monitor sections to understand their usage.

Then I created a linked service to connect ADF with Azure Blob Storage. I also created source and destination datasets for the CSV file.



Pipeline Creation

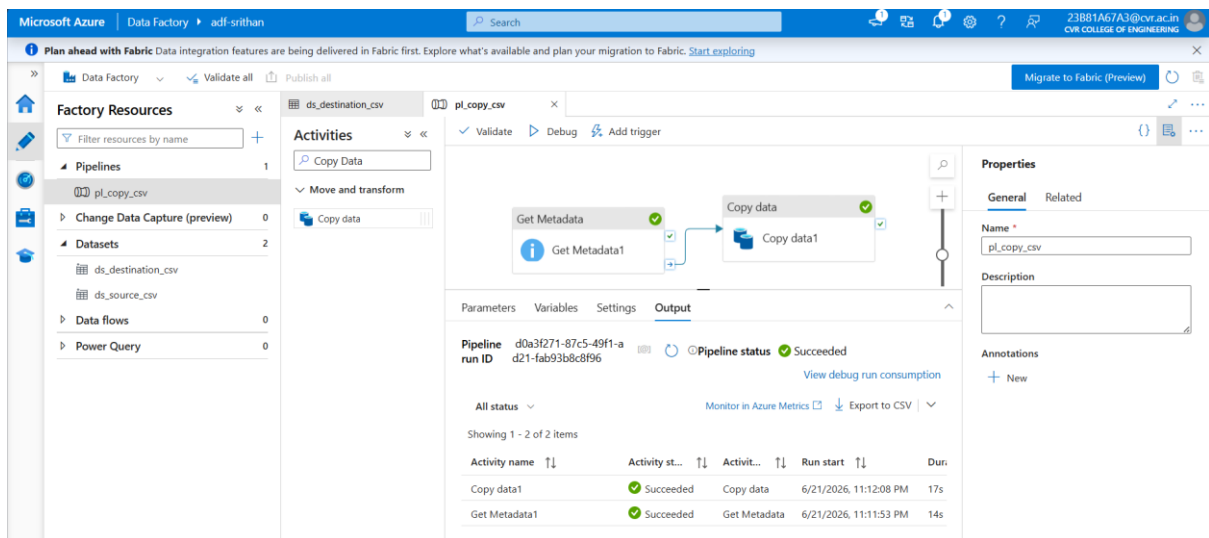
A pipeline named **pl_copy_csv** was created.

In the pipeline:

1. Get Metadata activity was used to check whether the source file exists and get file information.
2. Copy Data activity was used to copy data from the source container to the destination container.

Pipeline Execution

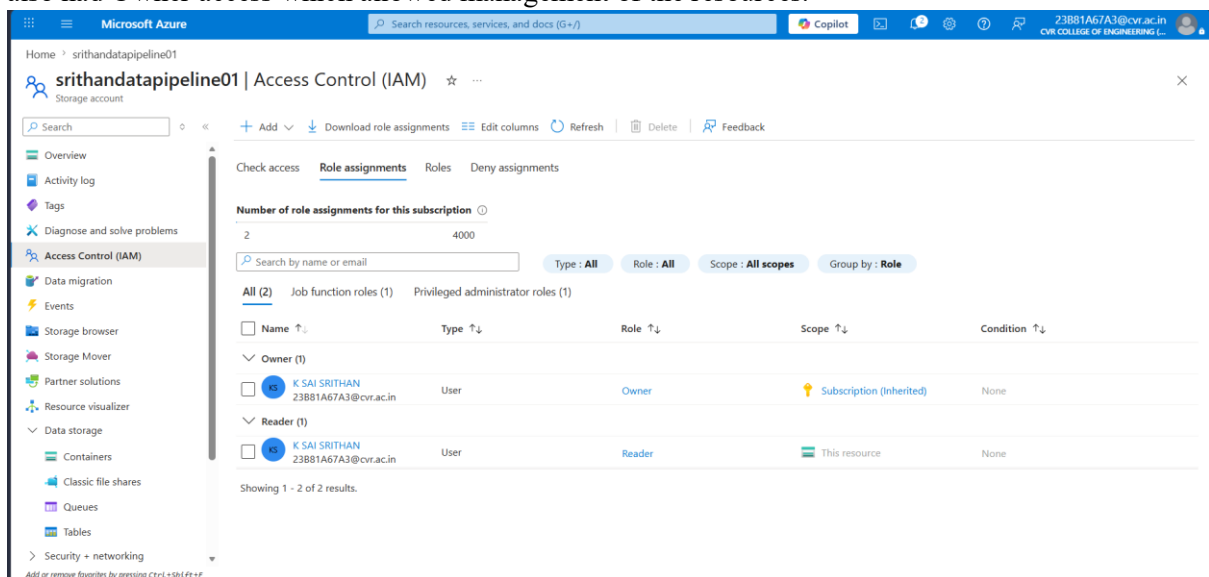
The pipeline was executed using the Debug option. Both activities completed successfully and the pipeline status showed **Succeeded**.



Activity name	Activity st...	Activit...	Run start	Dur...
Copy data1	✓ Succeeded	Copy data	6/21/2026, 11:12:08 PM	17s
Get Metadata1	✓ Succeeded	Get Metadata	6/21/2026, 11:11:53 PM	14s

IAM Roles

I checked the IAM settings of the storage account. Reader role was assigned successfully. The account also had Owner access which allowed management of the resources.



Name	Type	Role	Scope	Condition
K SAI SRITHAN	User	Owner	Subscription (Inherited)	None
K SAI SRITHAN	User	Reader	This resource	None

Pipeline Flow

Source CSV File (Blob Storage)



Get Metadata



Copy Data



Destination Blob Storage

Results

- Uploaded CSV file was detected successfully.
- File metadata was checked using Get Metadata activity.
- Data was transferred from source storage to destination storage.
- Pipeline execution was successful.
- Output file was generated in the destination container.

Conclusion

In this assignment, I learned how to use Azure Storage and Azure Data Factory to build a simple data pipeline. I created the required resources, connected them using linked services and datasets, validated the source file using Get Metadata, and copied the data successfully to another location using Copy Data activity.